

LECTURE 13

PROCURING GOODS AND SERVICES FOR IT

Knowledge Area	Process				
	Initiating	Planning	Executing	Monitoring & Control	Closing
Procurement Management		Plan Procurement Management	Conduct Procurements	Control Procurements	Close Procurements

What is Procurement?

Sometimes you need to hire an outside company to do some of your project work. That is called procurement, and the outside company is called the seller / vendors / Contractor.

Or

Procurement is the act of obtaining goods or services, typically for business purposes.

Procurement Management Processes

- Plan Procurement Management
- Conduct Procurements
- Control Procurements
- Close Procurements

Plan Procurement Management Process

Plan out what you will purchase, and how and when you will need the contracts to be negotiated for your project.

Tools & Techniques:

Make or buy analysis means figuring out whether or not you should be contracting the work or doing it yourself. It could also mean deciding whether to build your own solution to your problem or buy one that is already available. Most of the same factors that help you make every other major project decision will help you with this one.

Expert judgment means asking someone who's made the same kind of decision before to help you look at all the information you have for your project and make the right decision. Experts can be really helpful in evaluating technology, or providing insight into how your work might be done in different sourcing scenarios.

Market research you will want to check out reviews of possible vendors to work with. Sometimes procurement teams will go to conferences or read published reports that evaluate vendors doing similar contracts to help make decisions.

Meetings help your team put their heads together and make sure they are covering all of the project's needs when setting up procurement.

Types of Procurement Contract

Any contract between two or more parties to deliver services or goods is a procurement contract. You can divide procurement contracts into three categories:

1. Fixed-Price Contract
2. Cost Reimbursable Contract
3. Time and Materials Contract

As a project manager, it is your job to select a contract that satisfies your requirements.

Fixed-Price Contract

Many experts call a Fixed-Price contract a lump-sum contract. You use this contract when the scope of work is fixed. Once the contract is signed, the seller is contractually bound to complete the task within the agreed price and time. Therefore, the seller bears most of the risk. There is no price renegotiation unless the scope of work changes.

- Generally, outsourcing and turnkey procurement contracts are Fixed-Price. This contract is useful with a well-defined scope.
- Since the cost cannot change, this contract is suitable for controlling costs.
- Here, change in scope is a costly affair. Contractors often get the contract by bidding the lowest price and then try to generate extra revenue at any opportunity, such as an added scope.
- We have seen contractors quarrel with project managers regarding the scope. They agreed initially but later argue on small issues to raise the change request.
- Make sure the scope is well-defined and as detailed as possible; observe any changes in scope carefully.

You can divide a Fixed-Price contracts into three categories:

1. Firm-Fixed-Price contract (FFP)
2. Fixed Price Incentive Fee contract (FPIF)
3. Fixed Price with Economic Price Adjustment Contract (FP-EPA)

Firm Fixed-Price Contract (FFP)

This is the simplest type of procurement contract. The seller has to complete the job within an agreed amount of money and time. The seller is responsible for any increase in cost and they are legally bound to complete the task within the agreement.

A Firm-Fixed-Price contract is mostly used in government or semi-government contracts where the scope of work is defined with high detail.

This contract is easy to float and receive bids, and it is evaluated on a cost basis, which is fairly quick.

Since the seller bears the risk, the cost is higher. If the scope is not clear, the seller and buyer may have disputes. Any deviation from the original scope can cost you a lot.

Example: The seller has to paint the whole building for 50,000 USD within 18 months.

Fixed-Price Incentive Fee Contract (FPIF)

Although the price is fixed, the seller may receive an incentive if they perform well. This incentive lowers the seller's risk.

The incentive can be tied to any project metrics such as cost, time, or technical performance.

Example: The contractor will receive an incentive of 10,000 USD if they achieve the first milestone on time.

Fixed-Price with Economic Price Adjustment Contracts (FP-EPA)

You use a Fixed-Price with Economic Price Adjustment Contract when the agreement is multi-year. This contract has a special provision that protects the seller from inflation.

Example: About 3% of the cost of the project will increase after a certain time based on the Consumer Price Index.

Purchase Order (PO)

This contract is used to buy commodities.

Example: Buying 10,000 bolts of cloth at the cost of 1.00 USD

Cost Reimbursable Contract

Many experts call this contract a Cost Disbursable contract.

Here, the seller is reimbursed for completed work plus a fee representing their profit. Often, sellers get this fee if they meet or exceed the selected project objectives: completing the task earlier or saving costs, etc.

You use a Cost Reimbursable contract when there is uncertainty in the scope, or if the risk is higher. In this contract, the risk is on the buyer as they pay for all costs.

This contract provides you with better cost control when you do not have a well-defined scope.

Scope Creep is an inherent drawback of a Cost Reimbursement Contract when the requirements are unclear. The seller may try to increase the cost to increase the fee or reimbursement.

You can minimize this through proper monitoring or capping the profit. For example, the maximum profit is 10% of the total cost.

You can divide Cost Reimbursable contracts into four categories:

1. Cost Plus Fixed Fee (CPFF)
2. Cost Plus Incentive Fee (CPIF)
3. Cost Plus Award Fee (CPAF)
4. Cost Plus Percentage of Cost (CPPC)

Cost Plus Fixed Fee Contract (CPFF)

The seller is paid for all incurred costs plus a fixed fee, regardless of their performance. The buyer bears the risk.

Organizations use this contract with high-risk projects where bidders are not interested in competing.

CPFF contracts keep the seller safe from risks.

Example: Total cost plus 25,000 USD as a fee.

Cost Plus Incentive Fee Contract (CPIF)

In a Cost Plus Incentive Fee contract, the seller will be reimbursed for all costs plus an incentive fee based upon achieving certain performance objectives mentioned in the contract. This incentive will be calculated using an agreed-upon formula.

Here, the risk lies with the buyer; however, it is lower than in the Cost Plus Fixed Fee.

In a CPIF contract, the incentive is a motivating factor for the seller.

Generally, an incentive is a percentage of the savings that buyer and seller share.

Example: If the project is completed under budget, the seller will receive 25% of the savings.

Cost Plus Award Fee (CPAF)

Here, the seller is paid for their costs plus an award fee. This extra will be based on achieving satisfaction according to specified performance objectives described in the contract.

Note: There is a difference between an incentive fee and award fee. An incentive fee is calculated based on a formula defined in the contract and is an objective evaluation. An award fee depends on the client's satisfaction and is evaluated subjectively. An award fee is not subject to an appeal.

Example: If the seller completes the task by meeting or exceeding quality standards based on their performance, the buyer may give an award up to 10,000 USD.

Cost Plus Percentage of Cost (CPPC)

Here, the seller is paid for all costs incurred plus a percentage of these costs. Buyers often do not prefer this type of contract because the seller might artificially increase the costs to earn a higher profit.

Example: Total cost plus 15% of the cost as a fee to the contractor.

Time and Materials Contract

This is a hybrid of Fixed-Price and Cost Reimbursable contracts. Here, both parties share the risk.

You use a Time and Materials contract when the deliverable is “labor hours.” Here, the project manager will state the required qualifications and experience to the seller who will provide the staff.

This contract is used to hire experts or outside support.

In this contract, the buyer can specify the hourly rate with a “not-to-exceed” limit.

Example: You will pay a technician 20 USD per hour.

Point to remember: Even if your project has several contracts, they don't all have to be the same type. That's why you need to administer each one separately.

Conduct Procurements Process

Decide on the seller (or sellers) you are going to work with, and finalize and sign the contract.

As you **search** for your **vendors**, your process works a little something like this:

1. Send out a Request for Information (RFI).

2. Set up a bidders' conference.

A bidders' conference is optional, not required.

3. Send a Request for Proposals (RFP).

4. Review proposals and make your final decision.

Tools & Techniques:

Analytical techniques

You will need to determine whether or not the vendor is capable of completing the project in the required time frame and understand how well it can deliver to your budget. If you check out its track record on past projects, you'll have a better sense of how it'll perform.

Independent estimates

A lot of the time, you do not have the expertise in your company to figure out whether or not a seller is quoting you a fair price. That is why you will often turn to a third party to come up with an estimate of what the work should cost.

Expert judgment

Here is another case where it is really good to bring in someone from outside your project to help evaluate each proposal. You should bring in someone with a lot of specific expertise in the work being done to make sure the seller is up to the job.

What does bidder mean?

In a market, a **bidder** is a party offering to buy an asset from a seller at a specific price.

Bidder conferences

It is really important that you make sure all of the bidders can compete in a fair, unbiased way. And the best way to do that is to get them all in a room together, so that they can ask questions about your contract. That way, you don't give any one seller an advantage by providing inside information that the other sellers don't have access to.

Advertising

Sometimes the best way to get in touch with sellers is to place an ad. Also, sometimes you are **required** to take out an ad (like for some government-funded projects), in order to give all sellers full notice.

Procurement negotiation

When you send out a package of procurement documents to a potential seller, it usually has some information about the contract you want to sign: the type of contract, some of the terms, maybe

some rough calculations and estimates of the total costs, and other numbers. But not all sellers will want to sign that particular contract, even ones who you'll eventually want to work with. That's why you need to negotiate the terms of the contract.

Proposal evaluation techniques

You are going to have to work closely with the seller to figure out if his proposal really is appropriate for the work. You need to be very careful before you choose someone to do the work.

- **RFI: Request for information** is when a project owner has a problem they need fixed but do not know the best way to address it, they may issue a request for information (RFI) to gather possible remedies from contractors.
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- **RFP: Request for proposal** is when once the project owner has an idea of the project they would like to contract out, they can send out a request for proposal (RFP) from several contractors.
- **RFQ: Request for quotation** is related, but much more price-driven than the other two proposal types. The owner usually has selected the solution they want to use and just wants to know how much it will cost. There are not a lot of questions about qualifications to answer, as it is more about **pricing**, **payment options**, and **delivery timelines**.

Control Procurements Process

When the contract is under way, you stay on top of the work and make sure the contract is adhered to. You monitor what the contractor is producing and make sure everything is running smoothly. Occasionally, you will need to make changes to the contract. Here is where you will find and request those changes.

Tools and Techniques:

Payment systems

Your partner will not be very happy if you do not pay. The payment system is how your company pays its sellers. It is usually established by an Account or Accounts Receivable department.

Records management system

There are a lot of records produced by a typical contract: invoices, receipts, communications, memos, emails, instructions, clarifications, etc. You will need to put a system in place to manage them.

Inspection and audits

It means that you are looking at the products that the seller delivered to see if they meet your standards.

Close Procurements Process

When the work is done, you will close your contract out. You will make sure that the product that is produced meets the criteria for the contract, and that the contractor gets paid.